UDCT v7.12 Post-registration Results

Anisotropy and Screening-Length Boundaries of Signed Jeans Solutions in Hydrus I

Won Shik Paik
Independent Researcher | Auckland, New Zealand

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Results repository: github.com/wspaik5/UDCT_v7.12_Postregistration_Results

Abstract

We executed the frozen v7.12 follow-up protocol after screenshot evidence of public preregistration. The fixed Hydrus I unsoftened-cusp matrix comprises five screened GQUMOND model variants at six screening lengths and eleven constant anisotropies, with three QUMOND controls. All 198 fixed field evaluations and 2,178 pressure records completed, yielding 363 primary decisions: 274 negative (N), 48 positive sampled candidates (P), and 41 indeterminate (I). The screened subset gives N/P/I = 274/23/33; controls give 0/25/8.

All screened models are N at 2.5 and 3 pc throughout the tested anisotropy range. Some shorter-length, larger-anisotropy settings retain P candidates, with different extents across model variants. Of 575 prescribed adjacent intervals, 24 contain strict raw sign changes; all stop unresolved at endpoint or midpoint numerical margins. No requested coordinate-width target is reached. All 246 fixed-plus-adaptive isotropic reproduction gates pass.

What advanced

The experiment replaces a sparse sensitivity check with a complete registered sign map over $\beta=0$ to 0.5 and screening length $\ell=1$ to 3 pc. It identifies sampled regions excluded by this stationary closure, distinguishes the remaining positive candidates across models, and records the numerical uncertainty around transitions. This is progress in mapping conditional compatibility, not evidence of a viable replacement for dark matter.

Status and limits

No exact critical β or ℓ is established. Positive second moments do not establish a positive distribution function, equilibrium, dynamical stability or observational agreement. The experiment does not generally exclude MOND or validate a universal Hessian threshold. A subsequent Zenodo API retrieval confirmed the published record and its creation timestamp, 2026-10-10T12:09:24.904006Z. The original CLI used an observed-public-status upper bound; this historical provenance and the verified timestamps are distinguished on page 7.

1. Registered design and numerical rules

This is a prospective follow-up motivated by known v7.11 outcomes, not a blind prediction of those earlier cases. The original endpoints are rerun as anchors. Both fixed stages precede the adaptive search. All four scientific Python sources remain byte-identical to the preregistered package.

Item	Executed specification
Target and density	Hydrus I; unsoftened Tapered cusp, ε=0; inherited fixed stellar and Galactic-field inputs.
Screened models	BASE, A-f1/2, A-f2, A-Qs, A-Qd.
Anisotropy β	0, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50.
Screening length ℓ (pc)	1, 1.2, 1.5, 2, 2.5, 3.
QUMOND controls	nu0, nu1, nu2 at all eleven β values; no repeated length coordinate.
Numerical comparisons	smoke/default/refined grids; two inner limits, 0.002 and 0.0005 r_h.
Boundary and reporting	Outer pressure zero at 40 r_h; 1,200 logarithmic common radii from 0.002 to 10 r_h, with valid native minima retained.

Signed stationary Jeans closure

The solver integrates $dP_r/dr + 2\beta P_r/r = -\rho_\star g_{in}$, with $P_r=\rho_\star\sigma_r^2$ and positive inward acceleration g_{in}. Negative pressure is retained without clipping, absolute values or square-root conversion. Normalization is $v_{ref}f^2=r_h g_{N,\star}(r_h)$. Changing β changes Jeans weighting while the frozen field and Hessian for that model/length remain unchanged.

Conservative numerical classification

Let a,b,c,d denote normalized common variance arrays for default/0.002, refined/0.002, default/0.0005 and refined/0.0005. The empirical discrepancy E=max(sup|a-b|, sup|c-d|, sup|b-d|), and $\tau=\max(10^{-8},5E)$. This is a numerical diagnostic, not a rigorous continuum error bound.

N	At least one common radius has all four deciding values below -τ.
P	All four common arrays and their valid native minima exceed τ.
I	Every other outcome, including values within the margin or inconsistent comparisons.

Validation

Package hashes, pinned NumPy/SciPy versions and synthetic tests passed. Gate R reproduces inherited β=0 variance at every field evaluation to normalized error $\le 10^{-10}$; all 246 gates passed, with recorded maximum error 0.0. This checks solver consistency, not complete physical validity or convergence to the continuum solution.

2. Complete fixed-grid sign map

Each cell is one primary decision from six grid/inner-limit pressure records. Columns show β; ℓ is in pc. Colored labels retain uncertainty explicitly. Controls have no screening-length coordinate.

Model	ℓ	0.00	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
BASE	1	N	N	N	N	N	N	I	I	I	P	P
BASE	1.2	N	N	N	N	N	N	N	I	I	P	P
BASE	1.5	N	N	N	N	N	N	N	N	I	I	P
BASE	2	N	N	N	N	N	N	N	N	N	N	I
BASE	2.5	N	N	N	N	N	N	N	N	N	N	N
BASE	3	N	N	N	N	N	N	N	N	N	N	N
A-f1/2	1	N	N	N	N	N	N	N	N	N	N	I
A-f1/2	1.2	N	N	N	N	N	N	N	N	N	N	N
A-f1/2	1.5	N	N	N	N	N	N	N	N	N	N	N
A-f1/2	2	N	N	N	N	N	N	N	N	N	N	N
A-f1/2	2.5	N	N	N	N	N	N	N	N	N	N	N
A-f1/2	3	N	N	N	N	N	N	N	N	N	N	N
A-f2	1	N	N	N	N	N	I	I	I	I	P	P
A-f2	1.2	N	N	N	N	N	N	N	I	I	P	P
A-f2	1.5	N	N	N	N	N	N	N	N	N	I	P
A-f2	2	N	N	N	N	N	N	N	N	N	N	I
A-f2	2.5	N	N	N	N	N	N	N	N	N	N	N
A-f2	3	N	N	N	N	N	N	N	N	N	N	N
A-Qs	1	N	N	N	N	N	I	I	I	P	P	P
A-Qs	1.2	N	N	N	N	N	N	I	I	I	P	P
A-Qs	1.5	N	N	N	N	N	N	N	N	I	P	P
A-Qs	2	N	N	N	N	N	N	N	N	N	I	P
A-Qs	2.5	N	N	N	N	N	N	N	N	N	N	N
A-Qs	3	N	N	N	N	N	N	N	N	N	N	N
A-Qd	1	N	N	N	N	N	N	I	I	I	P	P
A-Qd	1.2	N	N	N	N	N	N	N	I	I	P	P
A-Qd	1.5	N	N	N	N	N	N	N	N	I	I	P
A-Qd	2	N	N	N	N	N	N	N	N	N	N	I
A-Qd	2.5	N	N	N	N	N	N	N	N	N	N	N
A-Qd	3	N	N	N	N	N	N	N	N	N	N	N
QUMOND-nu0	-	I	I	I	P	P	P	P	P	P	P	P
QUMOND-nu1	-	I	I	P	P	P	P	P	P	P	P	P
QUMOND-nu2	-	I	I	I	P	P	P	P	P	P	P	P

N: resolved negative common band. P: positive sampled second-moment candidate. I: indeterminate at the registered numerical margin. Counts are parameter-grid outcomes, not empirical success probabilities.

3. Results and the nature of the progress

Model	N	P	I	Total
BASE	53	5	8	66
A-f1/2	65	0	1	66
A-f2	53	5	8	66
A-Qs	50	8	8	66
A-Qd	53	5	8	66
QUMOND-nu0	0	8	3	11
QUMOND-nu1	0	9	2	11
QUMOND-nu2	0	8	3	11
Screened subtotal	274	23	33	330
Control subtotal	0	25	8	33
Fixed total	274	48	41	363

Length and anisotropy dependence

For all five screened models, every tested β is N at ℓ=2.5 and 3 pc. At ℓ=1 pc, BASE, A-f2 and A-Qd are P at β=0.45 and 0.50; A-Qs is P at β=0.40, 0.45 and 0.50. At ℓ=1.5 pc, BASE, A-f2 and A-Qd retain P only at β=0.50, while A-Qs also retains β=0.45. At ℓ=2 pc, only A-Qs retains a P case, at β=0.50.

A-f1/2 has 65 N cases and one I case (ℓ=1 pc, β=0.50); no P case is found. Its absence of sampled P is restricted to the registered model, profile, domain and numerical rules. QUMOND controls have no N cases, but eight control cases remain I; they cannot all be counted as positive.

Advance relative to the motivating v7.11 evidence

The inherited v7.11 protocol sampled β=-1, -0.5, 0, 0.25 and 0.5 and lengths 1, 3 and 10 pc, across cusp and Plummer profiles. v7.12 focuses on the unsoftened cusp and resolves the interval β=0 to 0.5 with eleven values and ℓ=1 to 3 pc with six lengths. It adds intermediate settings while preserving previously known anchors.

The advance is threefold: a complete parameter map under fixed assumptions; a measured distinction between model variants; and a transparent record of where numerical uncertainty prevents a precise transition claim. It narrows the sampled conditions requiring further study. It does not independently predict the already-known endpoints, establish a continuous admissible boundary, or show that a remaining P candidate is a realizable stellar system.

4. Adaptive refinement: all 24 outcomes

The registered scalar F is the lowest normalized variance over the four deciding common arrays and valid native minima. All 575 adjacent screened-model intervals were checked; 24 have opposite raw endpoint signs. Numerical signs require F below -τ or above τ. A raw sign change alone is insufficient to claim a resolved N-to-P transition.

Axis	Model	Fixed coordinate	Initial interval	Retained interval	Stop / steps
β	BASE	ℓ=1 pc	0.35 - 0.4	0.35 - 0.4	Endpoint / 0
β	BASE	ℓ=1.2 pc	0.4 - 0.45	0.4 - 0.45	Endpoint / 0
β	BASE	ℓ=1.5 pc	0.45 - 0.5	0.45 - 0.5	Endpoint / 0
ℓ (pc)	BASE	β=0.4	1 - 1.2	1 - 1.2	Endpoint / 0
ℓ (pc)	BASE	β=0.45	1.2 - 1.5	1.2 - 1.5	Endpoint / 0
ℓ (pc)	BASE	β=0.5	1.5 - 2	1.5 - 2	Endpoint / 0
β	A-f2	ℓ=1 pc	0.4 - 0.45	0.4 - 0.45	Endpoint / 0
β	A-f2	ℓ=1.2 pc	0.4 - 0.45	0.4 - 0.45	Midpoint / 1
β	A-f2	ℓ=1.5 pc	0.45 - 0.5	0.45 - 0.475	Midpoint / 2
ℓ (pc)	A-f2	β=0.45	1.2 - 1.5	1.2 - 1.5	Midpoint / 1
ℓ (pc)	A-f2	β=0.5	1.5 - 2	1.5 - 2	Endpoint / 0
β	A-Qs	ℓ=1 pc	0.35 - 0.4	0.35 - 0.4	Midpoint / 1
β	A-Qs	ℓ=1.2 pc	0.35 - 0.4	0.35 - 0.4	Endpoint / 0
β	A-Qs	ℓ=1.5 pc	0.4 - 0.45	0.4 - 0.45	Midpoint / 1
β	A-Qs	ℓ=2 pc	0.45 - 0.5	0.475 - 0.5	Midpoint / 2
ℓ (pc)	A-Qs	β=0.4	1.2 - 1.5	1.2 - 1.5	Endpoint / 0
ℓ (pc)	A-Qs	β=0.45	1.5 - 2	1.5 - 1.75	Midpoint / 2
ℓ (pc)	A-Qs	β=0.5	2 - 2.5	2 - 2.25	Midpoint / 2
β	A-Qd	ℓ=1 pc	0.35 - 0.4	0.35 - 0.4	Endpoint / 0
β	A-Qd	ℓ=1.2 pc	0.4 - 0.45	0.4 - 0.45	Midpoint / 1
β	A-Qd	ℓ=1.5 pc	0.45 - 0.5	0.45 - 0.475	Midpoint / 2
ℓ (pc)	A-Qd	β=0.4	1 - 1.2	1 - 1.2	Endpoint / 0
ℓ (pc)	A-Qd	β=0.45	1.2 - 1.5	1.2 - 1.5	Midpoint / 1
ℓ (pc)	A-Qd	β=0.5	1.5 - 2	1.75 - 2	Midpoint / 2

Endpoint: stopped before midpoint evaluation because at least one endpoint lacked numerical margin. Midpoint: stopped when an evaluated midpoint lacked margin. Steps count new midpoint evaluations. Every retained interval is unresolved; no row reached width ≤0.001 in β or ≤0.01 pc in ℓ.

Completion, not a computation failure

Twelve brackets stopped at endpoints and twelve at midpoints. The search obeyed its pre-specified stopping rules, adding 18 cases. Total counts are 246 fields, 2,286 pressure records and 381 decisions. No resource failure or iteration-limit termination occurred. The fixed map remains valid within its stated numerical scope; no precise critical coordinate is established.

A bracket can contain multiple roots; tangential zeros without sampled sign changes can be missed. The numerical lower-envelope zero is not necessarily a local acceleration-zero radius or a Hessian eigenvalue threshold.

5. Future directions and conclusion

Future directions - not executed in this report

The following proposals are motivated by these observed outcomes. They are not part of the completed v7.12 preregistered experiment and must be labeled as subsequent work. A new public specification should precede any confirmatory follow-up, including its grids, coordinates, tolerances, budgets and stopping rules.

Priority	Question and prospective test
1. Numerical resolution	Determine whether the I cases and unresolved brackets can be resolved. Predefine additional radial/angular grids, inner-limit checks, comparison norms and stopping limits. Refine both negative and positive neighboring cases without selecting only favorable outcomes. Report unresolved outcomes if the margin persists.
2. Distribution-function admissibility	For a pre-specified set of P candidates, formulate an appropriate distribution-function reconstruction under the adopted potential and constant anisotropy. Establish nonnegativity and the applicable assumptions before interpreting second moments as a realizable stellar system.
3. Stability	Only after identifying admissible candidates, specify the dynamical equations, perturbations, boundary conditions, diagnostics and time resolution for a stability test. A stationary Jeans calculation alone provides no such test.
4. Observational and cross-target checks	Predefine stellar inputs, uncertainties, projection and selection treatment, and observational comparison criteria. Test frozen predictions on additional targets. Separate any parameter estimation from independent validation.

Scope discipline

Core softening, variable β(r), changed external fields and local Hessian/u diagnostics require separate pre-execution definitions. They must not be introduced as silent repairs to this result. The present β scan does not alter the Hessian at fixed model and length, so it cannot establish that changing a Hessian eigenvalue restores physical viability.

Conclusion

v7.12 advances the reproducible characterization of signed Jeans solutions in Hydrus I. It confirms broad negative regions at longer tested screening lengths and retains a model-dependent set of positive sampled candidates at shorter lengths and larger anisotropies. All fixed cases and the prescribed adaptive search completed with unchanged scientific sources and passing reproduction gates.

The remaining uncertainty is substantive: none of the 24 raw sign-changing brackets achieves the registered coordinate precision. The justified conclusion is a sharper sampled map of conditional compatibility, with explicit unresolved boundaries. Physical admissibility, stability and observational success remain open. These results support targeted, separately specified follow-up tests; they do not establish general MOND exclusion or the physical validation of Hessian-dependent models.

6. Provenance, reproducibility and references

Execution record

Production started 10 October 2026 at 13:24:40.435703 UTC (11 October, 02:24:40 Auckland). Calculations completed at 13:25:01.785279 UTC; final result installation was logged at 13:25:11.985649 UTC. Environment: Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0. The complete compressed output reloaded, all fixed/total counts were checked, and its SHA-256 matched the success journal entry. One completed attempt is reported.

Verified Zenodo timestamps and original execution provenance

A subsequent direct retrieval from the public Zenodo records API confirmed status=published, state=done and submitted=true for record 23280972. Its **created** timestamp is **2026-10-10T12:09:24.904006Z** (11 October 2026, 01:09:24.904006 Auckland); its **modified** timestamp is 2026-10-10T12:09:27.242302Z (01:09:27.242302 Auckland). The created field is the record creation time, not a separately exposed publication-button event. The metadata publication_date is 2026-10-11 and is not a clock timestamp. Production started **1 hour 15 minutes 15.531697 seconds after the recorded creation time**. The later API response corroborates the pre-execution Published/Open screenshots.

At execution, initial network retrievals had failed. The unchanged runner was therefore supplied registration_published_utc=2026-10-10T13:24:19Z, an observed-public-status upper bound, not an exact publication time. The original result and journal retain that value and their original SHA-256; they have not been rewritten or rerun. The departure from the README timestamp instruction remains disclosed, while the prior statement that independent network verification was unavailable is superseded by the successful later API retrieval. No scientific source, numerical result, classification or conclusion changes.

API evidence: https://zenodo.org/api/records/23280972. Field definition: Zenodo search guide (created: record creation timestamp). Timing provenance revised 11 October 2026 (Auckland).

Result and source checksums (SHA-256)

Complete result: 2ed68841f3ada9427b9200e5f8c05f0c
743af5dc60c131306643ed3e84b7be54

v7.12 runner: 3b5f93766940243d257f865691f708fd
904b2ba9384003ae407f911179059b49

v7.11 signed solver: 3f00abae251d7be64a4ca5abb48b4f55
0644b21c946206571fd448464b13ade2

v7.10 gravity: a4d1a07ebcaa06348be3022d0f1f6ca4
63347e00194f7fde59f2bd9dbcd521b7

v7.06 antecedent: a64440e1350ae8bf10eb3188ec34829b
6d586abf275de5cdc841a50ad3478284

The lightweight upload package includes unchanged scientific Python sources, an execution launcher, all case and bracket CSVs, console log and timestamp evidence. Full original compressed output and incremental journal are retained separately; they are not included in this code-and-tables package. The case CSV includes 363 primary and 18 adaptive cases; use stage1/stage2 for primary counts. Historical preparation-status text in the frozen source is retained, not a statement that this completed run preceded publication.

Primary records and links

[1] Paik, W. S. (2026). UDCT v7.12 Preregistration: Anisotropy and Screening-Length Boundaries of Signed Jeans Solutions in Hydrus I. https://doi.org/10.5281/zenodo.23280972

[2] Paik, W. S. (2026). UDCT v7.11 Post-registration Results: Constant-Anisotropy Jeans and Outer-Boundary Sensitivity in Hydrus I. https://doi.org/10.5281/zenodo.23263099

[3] Paik, W. S. (2026). UDCT v7.11 Preregistration. https://doi.org/10.5281/zenodo.23244522

[4] Paik, W. S. (2026). UDCT v7.10 Post-registration Results. https://doi.org/10.5281/zenodo.23233837

Registered source repository: github.com/wspaik5/UDCT_v7.12_Preregistration_Jeans_Transitions

Results DOI (reserved until publication): https://doi.org/10.5281/zenodo.23296444. Results repository: github.com/wspaik5/UDCT_v7.12_Postregistration_Results. The preregistration DOI identifies the protocol separately. Prepared 11 October 2026 (Auckland).